
Micro-Internship Week 1: Evaluating the use of LLM-Engineering and Reasoning to improve on human-engineered features

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Abstract

We evaluate founder-success prediction using two complementary experiments: a Human vs LLM-Eng vs LLM-Reasoning experiment focused on the core methodology, and a model-testing experiment that examines pruning, feature representations, transforms including partial least squares (PLS), and model classes. The report consolidates validation and private-test performance, identifies generalization gaps, and motivates a final model suite for test-time prediction.

1. Introduction

Founder and early-stage company characteristics are noisy and high-dimensional, but investment decisions require interpretable, robust, and precise signals. This report consolidates the results of two experiments: (1) a Human vs LLM-Eng vs LLM-Reasoning experiment that defines the core method and evaluation protocol, and (2) a model-testing experiment that probes feature sets, pruning, transforms, and model classes.

2. Method and Results (Human vs LLM-Eng vs LLM-Reasoning)

2.1. Method

The Human vs LLM-Eng vs LLM-Reasoning experiment evaluates whether LLM-derived features improve founder-success prediction relative to manual baselines. Three feature families were compared: *Human*, *LLM-Eng*, and *LLM-Reasoning*. Human features were hand-crafted from exploratory analysis, LLM-Eng features were structured features generated from a small training subset passed through a single API call, and LLM-Reasoning features were subjective prompted outputs, requiring an API call for each

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Family	Validation $F_{0.5}$ (mean $\pm$ std)	Test $F_{0.5}$ (mean $\pm$ std)
Human	0.227 $\pm$ 0.002	0.2567 $\pm$ 0.0032
LLM-engineered	0.254 $\pm$ 0.005	0.2727 $\pm$ 0.0196
LLM-Reason (A, E)	0.277 $\pm$ 0.002	0.2028 $\pm$ 0.0605

Table 1. Part 1 family-level validation vs private-test.

founder, intended to capture signal not already encoded in the engineered sets.

The empirical design consisted of two experiments. The first was a 3x3 comparison of Human, LLM-Eng, and LLM-Reasoning variants using XGBoost only. Within this block, the three human variants are denoted *Human-1/2/3*, the three engineered variants are denoted *LLM-Eng-05/08/07*, and reasoning augmentation is denoted by *LLM-Reason (A, E)*. The second block evaluated reasoning additions to a high-quality human-engineered feature-set, denoted *HQ* - a human-engineered benchmark that served as the strongest manual baseline at this stage of the project. Each reasoning prompt family (A–F) was evaluated separately on the dataset to generate its own feature block. Labels such as *A*, *E* or *A, D, E, F* therefore denote that the corresponding prompt-specific feature blocks were concatenated after feature extraction and before model training, rather than produced by a single combined prompt. The combinations chosen to be evaluated on the test set were *HQ*, *HQ* with *D*, *HQ* with *A* and *E*, and *HQ* with *A, D, E*, and *F*, using both Logistic Regression (LR) and XGBoost with depth-1 trees, denoted *XGB1*. Validation performance is reported as mean, $\pm$, standard deviation across folds, while private-test performance is computed on held-out success labels.

2.2. Results

The first question is whether LLM-derived features improve on the human baseline once generalization is evaluated directly. Table 1 gives a mixed answer. At the family level, LLM-Eng improves on Human on both validation and private test, but adding LLM-Reasoning changes the pattern: validation performance rises further, while private-test performance deteriorates sharply.

Table 2 shows that this pattern is not an artifact of aggregation. Among the non-reasoning models, *LLM-Eng-05*

Model	Validation $F_{0.5}$ (mean $\pm$ std)	Test $F_{0.5}$
Human-1 (scaled durations)	0.226 $\pm$ 0.014	0.2530
Human-2 (binary equivalents)	0.226 $\pm$ 0.015	0.2607
Human-3 (mixed durations)	0.230 $\pm$ 0.016	0.2563
LLM-Eng-05	0.261 $\pm$ 0.066	0.2969
LLM-Eng-08	0.252 $\pm$ 0.042	0.2722
LLM-Eng-07	0.249 $\pm$ 0.051	0.2490
LLM-Eng-05 + LLM-Reason (A, E)	0.280 $\pm$ 0.069	0.1212
LLM-Eng-08 + LLM-Reason (A, E)	0.274 $\pm$ 0.032	0.2210
LLM-Eng-07 + LLM-Reason (A, E)	0.277 $\pm$ 0.040	0.2661

Table 2. Part 1 per-model validation vs private-test.

(a) Logistic Regression validation vs private-test for HQ and reasoning combinations.

Feature combo	LR val	LR test
HQ	0.234 $\pm$ 0.041	0.2684
HQ, D	0.249 $\pm$ 0.021	0.2482
HQ, A, D, E, F	0.339 $\pm$ 0.043	0.2243
HQ, A, E	0.281 $\pm$ 0.042	0.2225

(b) XGB1 validation vs private-test for HQ and reasoning combinations.

Feature combo	XGB1 val	XGB1 test
HQ	0.233 $\pm$ 0.027	0.2737
HQ, D	0.235 $\pm$ 0.029	0.2638
HQ, A, D, E, F	0.301 $\pm$ 0.042	0.2604
HQ, A, E	0.280 $\pm$ 0.055	0.2582

Table 3. Validation vs private-test for HQ benchmark with reasoning combinations.

is the strongest model on the private test set, outperforming all three Human variants. By contrast, the reasoning-augmented variants are unstable: *LLM-Eng-05 + LLM-Reasoning(A, E)* attains the strongest validation score in this block, but collapses on the private test set. This indicates that automated feature engineering is useful, whereas subjective reasoning features require much tighter control to avoid overfitting.

Having established that richer reasoning features can raise validation performance while weakening generalization, the second block asks whether the same pattern persists when reasoning combinations are added to the HQ benchmark, and if these generalise to the test set.

Table 3 reports the validation and private-test results for the HQ reasoning block, with table (a) showing LR and table (b) showing XGB1. In both cases, more complex reasoning combinations improve cross-validated performance relative to HQ, whereas private-test performance deteriorates with the addition of reasoning features.

3. Discussion (Human vs LLM-Eng vs LLM-Reasoning)

The Human vs LLM-Eng vs LLM-Reasoning results support two linked conclusions. First, LLM-Eng features match and sometimes improve on both the Human variants and the HQ baseline. Second, the apparent gains from LLM-Reasoning features are misleading when judged only by cross-validation. In both experiment blocks, reasoning combinations improve validation performance, but those gains do not carry through to the private test set. The most plausible explanation, later investigated directly in section 4, is that the added reasoning features introduce highly collinear parameters that increase flexibility without improving generalisation. The implication is not a lack of predictive signal, but a lack of stability once richer reasoning features are introduced.

4. Method and Results (Model Testing)

4.1. Method

The model-testing experiment was designed to answer a different question from the Human vs LLM-Eng vs LLM-Reasoning experiment: not whether richer feature families can produce high validation scores, but why cross-validated improvement failed to translate into better private-test performance, and which modeling choices remain stable enough to justify final test-set prediction. The experiment therefore evolved through a sequence of controlled investigations. First, regularisation and pruning were examined through ElasticNet-guided sweeps, which showed that aggressive pruning reduced variance and improved stability; this became the default pre-processing choice for later experiments. Under aggressive pruning, the following HQ features were removed from the original high-quality human baseline: *best_degree_prestige*, *comfort_index*, *longest_founding_tenure*, *persistence_score*, *has_prior_ipo*, *is_serial_founder*, *founding_role_count*, and *edu_prestige_tier*. This step was motivated by the hypothesis that redundant human-engineered features were contributing to the same instability observed when reasoning features were added.

Second, feature transformations were evaluated. Three representations were considered: the untransformed feature space (*None*), Principal Component Analysis (*PCA*), and Partial Least Squares (*PLS*). *PCA* was ultimately rejected because the objective here is not to maximize explained variance, but to preserve predictive signal in a lower-dimensional orthogonal representation. *PLS* is better aligned with that objective because it constructs components using covariance with the target. Third, model classes were compared: Logistic Regression (*LR*), XGBoost with depth-1 trees (*XGB1*), and multilayer perceptrons with 2, 4, and 32

hidden units, denoted *MLP2*, *MLP4*, and *MLP32*, respectively. Finally, co-linearity diagnostics based on variance inflation factors (VIF), correlations, and condition numbers were introduced to determine when dimensionality reduction should be used for stability rather than raw compression.

4.2. Results

Transform sweeps showed that PCA did not provide the right inductive bias for this task, whereas PLS yielded more reliable gains on high-signal feature combinations. The PLS sweep further showed that $n_{components} = 6$ offered the strongest overall tradeoff across LR and XGB1, with mean CV $F_{0.5}$ values of 0.273 and 0.270, respectively. This is why the final evaluation retained only BASE and PLS representations.

Sweeping regularisation for each model then clarified the appropriate level of functional complexity. LR remained the strongest stability anchor throughout the experiment, while XGB1 was competitive but often performed worse on CV with a larger variance. The neural models revealed a clear gradient: MLP32 had the highest tendency to overfit, MLP2 was comparatively stable but weak, and MLP4 provided the best balance of capacity and control, particularly under PLS. This demonstrated that the additional non-linearity is useful only when paired with a representation that has already reduced redundancy.

Aggressive pruning reduced redundancy, but not uniformly across feature combinations. Collinearity diagnostics showed that combinations involving A remained in a comparatively safe regime, whereas combinations built around F remained highly correlated. PLS was therefore not used universally: for cleaner combinations, BASE preserves interpretability without an obvious penalty, whereas for high-signal, high-collinearity combinations, PLS improves stability by compressing the redundant feature space into a smaller predictive representation.

The suite of models used on the test dataset was designed to separate three questions that would otherwise be conflated: whether the useful signal is clean or distributed, whether compression helps more than raw representation, and whether non-linear capacity adds value beyond a well-regularised linear model.

Table 4 brings these decisions together in the final model suite. It is organized by model class, transform, and reasoning-feature combination, making the design logic explicit: clean baselines are retained in the untransformed representation, while higher-signal combinations are evaluated under PLS.

Model	Transform	Reasoning features	CV $F_{0.5}$ (mean $\pm$ std)	Test $F_{0.5}$
LR	PLS	D, E, F	0.301 $\pm$ 0.036	0.274
LR	PLS	F	0.278 $\pm$ 0.031	0.271
MLP4	PLS	C, D, E, F	0.310 $\pm$ 0.044	0.270
LR	None	None	0.241 $\pm$ 0.033	0.264
LR	None	A, C	0.292 $\pm$ 0.033	0.242
LR	None	A	0.286 $\pm$ 0.028	0.240
MLP4	PLS	A, B, C, D, E, F	0.333 $\pm$ 0.042	0.239
LR	PLS	A	0.287 $\pm$ 0.036	0.233
LR	PLS	A, B, C, D, E, F	0.337 $\pm$ 0.033	0.225

Table 4. Final model suite (validation and private-test).

5. Discussion (Model Testing)

The PLS transformed LR model - with features F and D,E,F - remains comparatively stable from validation to private test, suggesting that PLS successfully compressed the feature space into a stable signal. However, these models do not clearly improve on the HQ baseline.

The full A, B, C, D, E, F combination was retained despite strong overfitting suspicions because it provided the highest cross-validated score and therefore served as a necessary stress test of whether maximum in-sample signal would generalise. The weaker private-test performance of many reasoning combinations confirms that high validation performance was likely due to ‘overfitting’ or noise created by redundant reasoning features. These results suggest that reasoning prompts A-F are not providing a clear signal.

6. Conclusions

The two experiments support a consistent conclusion. LLM-Eng features can improve on human baselines, but unrestricted reasoning features are prone to overfitting when judged on genuine hold-out performance. The model-testing experiment therefore shifts the selection criterion from validation maximization to stability under genuine distribution shift: aggressive pruning reduces redundancy, PLS is applied where co-linearity remains structurally high, and final model choice is restricted to specifications that preserve interpretable trade-offs between signal, compression, and complexity. The strongest evidence for this strategy is that PLS with F remains relatively stable between validation and private test, whereas the full A, B, C, D, E, F specification—retained because it achieved the best cross-validated score—does not generalise as well.

These results suggest that, despite the mitigation of co-linearity issues through the use of PLS, the LLM-Reasoning features did not provide a tangible improvement in model performance. We therefore suggest more sophisticated prompting techniques, to improve reasoning feature quality, may provide the largest uplift in model performance moving forward.

Impact Statement

This report aims to improve transparency and reliability in early-stage founder evaluation. Findings are limited to the observed dataset and should not be construed as investment advice.

Appendices

A. GitHub Repository

Python scripts for the experiments and pipelines described in this report is available at: https://github.com/tick-tock-code/Vela_internship_TRL.git

B. LLM Reasoning Prompts

B.1. Fixed Opening Prompt

```
You are analyzing a startup founder
profile.
Your goal is to extract signals that may
correlate with startup success.
IMPORTANT:
- Use only the provided information
- Be consistent across evaluations
- Avoid speculation beyond evidence
GENERAL PRINCIPLES:
- Do NOT rely primarily on university
  prestige or company brand names
- Focus on trajectory, responsibility, and
  evidence of ownership
- Look for patterns that are difficult to
  quantify with simple rules

Global output (always include in the final
JSON):
- evidence_support_rating (int 1-5)
  Description: After producing the scores,
  rate how well-supported the scores are
  by observable evidence.

{{EXPERIMENT_INSTRUCTIONS}}

Founder profile:
{{FOUNDER}}
```

Output ONLY valid JSON and include ALL required keys.

B.2. Prompt A

```
Score the following from 1 to 5:

- trajectory_strength
- ownership_signal
- career_coherence
- scrappiness

For each score, provide a one-sentence
justification.
```

```
Output keys:
- trajectory_strength
- trajectory_strength_justification
- ownership_signal
- ownership_signal_justification
- career_coherence
- career_coherence_justification
- scrappiness
- scrappiness_justification
```

```
Output format:
{
  "trajectory_strength": int,
  "trajectory_strength_justification":
    "...",
  "ownership_signal": int,
  "ownership_signal_justification": "...",
  "career_coherence": int,
  "career_coherence_justification": "...",
  "scrappiness": int,
  "scrappiness_justification": "..."
}
```

B.3. Prompt B

```
Use the following rubric:

Top founder:
- rapid progression or early leadership
- ownership in ambiguous environments
- operates in high-growth or technical
  contexts

Weak founder:
- limited ownership
- slow progression
- inconsistent direction

Score relative to this rubric (1-5).
  Provide a one-sentence justification
  for the score.

Output keys:
- rubric_score
- rubric_score_justification

Output format:
{
  "rubric_score": int,
  "rubric_score_justification": "..."
}
```

B.4. Prompt C

```
Identify 3 signals that are NOT directly
measurable
(e.g. not degree, company name, or years
of experience).

Then score:
- latent_strength: 1-5
- latent_risk: 1-5
```

Provide a one-sentence justification for each score that explicitly mentions the 3 signals.

Output keys:

- latent_strength
- latent_strength_justification
- latent_risk
- latent_risk_justification
- signal_1
- signal_2
- signal_3

Output format:

```
{
  "latent_strength": int,
  "latent_strength_justification": "...",
  "latent_risk": int,
  "latent_risk_justification": "...",
  "signal_1": "...",
  "signal_2": "...",
  "signal_3": "..."}
}
```

B.5. Prompt D

List:

- top_3_strengths (categorical labels)
- top_3_weaknesses (categorical labels)

Then, considering these, score:

- overall_founder_quality: 1-5

Provide a one-sentence justification that explicitly mentions the top_3_strengths and top_3_weaknesses.

Output keys:

- top_3_strengths
- top_3_weaknesses
- overall_founder_quality
- overall_founder_quality_justification

Output format:

```
{
  "top_3_strengths": "...",
  "top_3_weaknesses": "...",
  "overall_founder_quality": int,
  "overall_founder_quality_justification": "..."}
}
```

B.6. Prompt E

What aspects of this founder would be underrated or misleading if judged only by traditional metrics?

Then score:

- hidden_upside: 1-5
- hidden_risk: 1-5

Provide a one-sentence justification for each score that explicitly mentions the underrated/misleading aspects.

Output keys:

- hidden_upside
- hidden_upside_justification
- hidden_risk
- hidden_risk_justification
- underrated_aspects

Output format:

```
{
  "hidden_upside": int,
  "hidden_upside_justification": "...",
  "hidden_risk": int,
  "hidden_risk_justification": "...",
  "underrated_aspects": "..."}
}
```

B.7. Prompt F

Evaluate each of the following dimensions on a scale from 1 to 5, based solely on the founder's profile, using clear, objective, and unbiased descriptors for each. Feel free to rephrase the labels to minimize bias while maintaining their core intent:

- Growth Progression and Momentum
- Degree of Ownership and Accountability
- Career Focus and Consistency
- Resourcefulness and Flexibility in Challenging Situations

After assigning scores to these dimensions, determine an overall founder quality rating according to this criteria:

Top founder:

- Demonstrates rapid or sustained growth, or holds early leadership roles
- Exhibits strong ownership and responsibility in uncertain or ambiguous contexts
- Has experience within high-growth or technically complex environments

Weak founder:

- Shows limited ownership or responsibility
- Exhibits slow, stagnant, or minimal progression
- Lacks consistent focus or clear direction

In cases where evidence is mixed or ambiguous, assign a middle score and prioritize the most compelling evidence from the profile.

Provide a one-sentence justification for each score, including the overall founder quality rating.

Output keys:

- trajectory_strength
- trajectory_strength_justification
- ownership_signal
- ownership_signal_justification
- career_coherence
- career_coherence_justification
- scrappiness
- scrappiness_justification
- rubric_score
- rubric_score_justification

Output format:

```
{
  "trajectory_strength": int,
  "trajectory_strength_justification":
    "...",
  "ownership_signal": int,
  "ownership_signal_justification": "...",
  "career_coherence": int,
  "career_coherence_justification": "...",
  "scrappiness": int,
  "scrappiness_justification": "...",
  "rubric_score": int,
  "rubric_score_justification": "..."
}
```